

REDEVELOPMENT OF RANI KAMALAPATI RAILWAY STATION INTO A MULTIMODAL TRANSPORT HUB

**Integrating Railway, Metro, City Bus, Passenger Plaza and Underground
Parking**

Bachelor of Technology
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What is a Multimodal Transport Hub?

Over the last few decades, cities around the world have experienced rapid urban growth. As more people move to urban areas for better education, employment, and living opportunities, the demand for efficient transportation has increased significantly. Every day, thousands of commuters travel using different modes of transport such as trains, metro systems, buses, taxis, private vehicles, and two-wheelers. During peak hours, this often results in traffic congestion, overcrowded public transport, and delays, making urban mobility one of the biggest challenges faced by modern cities.

To overcome these challenges, cities are increasingly focusing on integrated transportation systems that allow people to travel more conveniently. This has led to the development of **Multimodal Transport Hubs (MTHs)**. A multimodal transport hub is a place where different modes of transportation are connected within a single, well-planned facility. Instead of functioning independently, railway stations, metro systems, city buses, taxis, auto-rickshaws, private vehicles, bicycles, and pedestrian pathways are brought together to form one integrated transport network. The main objective is to enable passengers to switch easily from one mode of transport to another with minimum walking distance, waiting time, and inconvenience.

The success of any public transportation system depends not only on the efficiency of individual transport modes but also on how well they are connected. A well-designed multimodal transport hub improves this connectivity by providing safe, direct, and convenient transfer facilities. It simplifies passenger movement through clear circulation paths, proper signage, barrier-free access, and organized transport zones. As a result, travel becomes faster, safer, and more comfortable for passengers.

Modern multimodal transport hubs are much more than transportation facilities. They also serve as important public spaces where people can shop, dine, wait comfortably, or simply move through the city with ease. Many successful transport hubs include passenger plazas, retail stores, food courts, waiting lounges, commercial spaces, landscaped open areas, and other public amenities. These features enhance the travel experience while also contributing to the economic and social development of the surrounding area.

The design of multimodal transport hubs differs from one city to another depending on local transportation systems and urban development. In many developed countries, these hubs are centered around railway stations that integrate metro services, buses, cycling infrastructure, and pedestrian networks. In developing countries like India, the transportation system is more diverse and often includes buses, metro rail, taxis, auto-rickshaws, ride-hailing services, private vehicles, and informal transport. Therefore, transport hubs in such cities must be carefully planned to accommodate all these modes while ensuring smooth traffic flow and efficient passenger movement.

Today, multimodal transport hubs play a vital role in creating sustainable and well-connected cities. By encouraging the use of public transport, reducing dependence on private vehicles, improving land utilization, and enhancing passenger convenience, they contribute to cleaner, safer, and more efficient urban environments. They are no longer viewed simply as transit

points but as key elements of modern urban planning that improve mobility, strengthen economic activity, and enhance the overall quality of life.

Example of interchange hubs and their specific issues

Manila: improving safety in a saturated, dangerous interchange hub



Aerial view of Bicutan station with passengers boarding and alighting

The offer of public transport available for the 12 millions inhabitants of the Filipino capital consists of a light rail transit system (three lines with a cumulated length of 46km), classic buses on main arterial roads, and a multitude of jeepneys – 40% of trips are made in these minibuses. All three of these means of transport come together at Bicutan station. Pedestrians can access bus lines and jeepneys by taking a footbridge over a high-traffic avenue, then crossing the railway. Street vendors are active on the roads.

Which Purposes Do Interchange Hubs Serve?

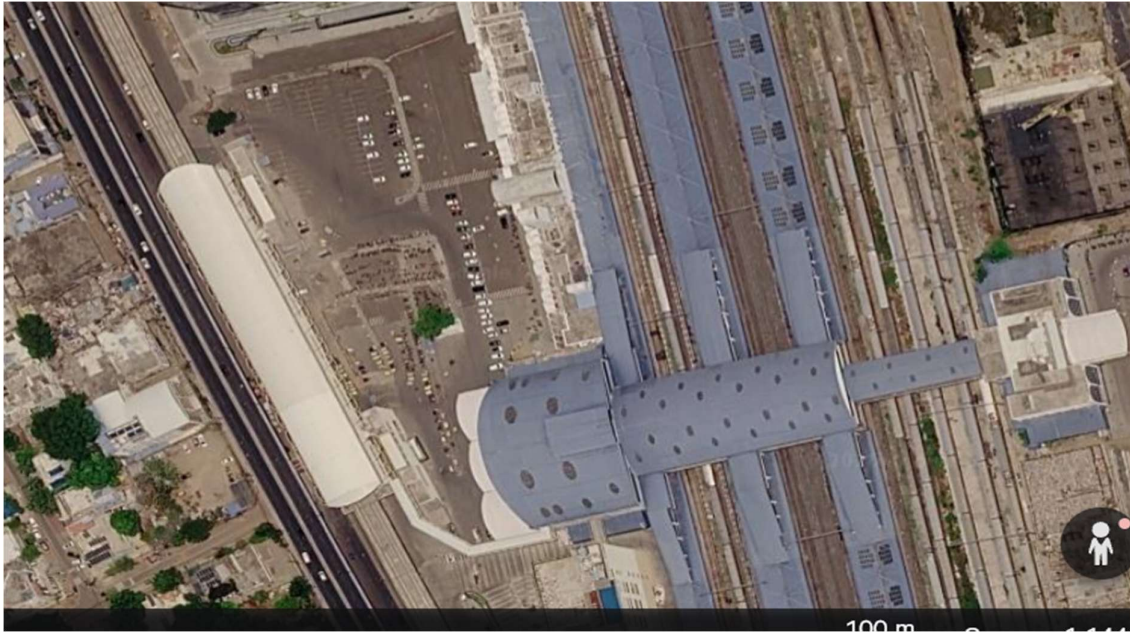
A multimodal transport hub is much more than a place where passengers change from one mode of transport to another. It plays an important role in improving urban mobility, enhancing the surrounding environment, and providing services that make travel more convenient and comfortable. In general, a multimodal transport hub serves three major purposes: transportation, urban integration, and passenger services.

The primary purpose of an interchange hub is to ensure the smooth movement of passengers between different modes of transport. By integrating facilities such as railways, metro systems, city buses, taxis, auto-rickshaws, and private vehicles, the hub allows passengers to transfer quickly and safely with minimal walking distance and waiting time. Efficient circulation planning, clear signage, and well-organized transport zones help reduce congestion and improve the overall efficiency of the transportation network.

Beyond transportation, an interchange hub also contributes to the development of the surrounding urban area. A well-designed hub connects different parts of the city, improves accessibility, and blends seamlessly with the existing urban fabric. It reduces physical barriers between transport systems and nearby neighborhoods while creating opportunities for redevelopment of underutilized land. Public plazas, landscaped spaces, pedestrian-friendly streets, and improved connectivity can transform the station area into an attractive and vibrant urban destination rather than simply a transit point.

Another important function of a multimodal transport hub is the provision of passenger services and public amenities. Modern transport hubs are designed to make travel more comfortable by providing facilities such as waiting lounges, retail outlets, restaurants, ticketing centres, information kiosks, restrooms, and seating areas. These services benefit not only passengers but also the surrounding community by creating active public spaces that support commercial activities and improve the overall quality of the urban environment.

The relative importance of these three functions may vary depending on the location, size, and transportation needs of a particular city. However, a successful multimodal transport hub always aims to balance efficient transportation, thoughtful urban planning, and high-quality passenger services to create a safe, accessible, and user-friendly mobility environment.



Need for the Project

Rani Kamalapati Railway Station is one of India's most modern railway stations and serves as an important transportation gateway for Bhopal and the surrounding region. Equipped with advanced passenger amenities, contemporary architecture, efficient railway operations, and direct connectivity to the metro network, the station has set a high standard for railway infrastructure in the country. It has significantly improved the travel experience for passengers and reflects the growing emphasis on modern and sustainable transportation facilities.

However, as passenger numbers continue to increase and cities become more dependent on integrated transportation systems, new challenges have emerged around the station precinct. Today's passengers rarely rely on a single mode of transport. A typical journey often involves a combination of trains, metro services, city buses, taxis, auto-rickshaws, ride-hailing services, private cars, or two-wheelers. While the railway station and metro are well connected, the surrounding forecourt has not evolved to accommodate this growing demand for seamless multimodal connectivity.

At present, a large portion of the station forecourt is occupied by surface parking. Although this serves the needs of private vehicle users, it also consumes valuable urban land that could be better utilized for public spaces and integrated transport facilities. The coexistence of pedestrians and vehicles within the same area creates congestion during peak hours, reduces passenger safety, and affects the overall efficiency of movement within the station premises. With passenger traffic expected to rise further in the future, the existing arrangement may not be sufficient to support the station's long-term transportation needs.

Another major limitation is the absence of a dedicated City Bus Terminal integrated with the railway and metro systems. At present, buses, taxis, auto-rickshaws, and app-based mobility services operate independently, forcing passengers to travel longer distances when transferring between different modes of transport. This lack of coordination reduces travel

convenience and prevents the station from functioning as a fully integrated multimodal transport hub.

The station precinct also offers an excellent opportunity to create high-quality public spaces that enhance the overall passenger experience. Around the world, successful multimodal transport hubs have shown that transportation infrastructure can also serve as vibrant public spaces with commercial, recreational, and social activities. Features such as pedestrian plazas, landscaped open areas, retail outlets, organized circulation routes, and clear wayfinding systems not only improve passenger comfort but also increase the urban and economic value of the surrounding area.

Considering these opportunities, this project focuses on redeveloping the station forecourt rather than modifying the existing railway station itself. The proposal retains the current railway and metro infrastructure while transforming the surrounding parking-dominated area into a modern multimodal transport hub. The concept includes relocating parking facilities underground, introducing a dedicated City Bus Terminal, developing a spacious passenger plaza, improving pedestrian circulation, and integrating commercial and public amenities into a single, well-planned transportation environment.

The proposed redevelopment demonstrates how underutilized station forecourts can be transformed into efficient urban mobility spaces. By improving multimodal connectivity, reducing surface congestion, making better use of available land, and creating a more passenger-friendly environment, the project aims to establish Rani Kamalapati Railway Station as a model for future multimodal transportation hubs in India.

Project Vision

The vision of this project is to transform the forecourt of Rani Kamalapati Railway Station into a modern multimodal transport hub that brings together different modes of transportation within a single, well-planned and passenger-friendly environment. The proposal goes beyond improving the railway station alone; it aims to create a vibrant urban space where transportation, public amenities, commercial activities, and open spaces work together to provide a seamless travel experience.

A key principle guiding this project is **"Preserve Existing Infrastructure, Enhance Passenger Experience."** Instead of reconstructing the existing railway station or metro infrastructure, the proposal builds upon these well-established facilities by improving the surrounding station precinct. The objective is to create a transport hub where passengers can move safely and conveniently between trains, metro services, city buses, taxis, auto-rickshaws, ride-hailing vehicles, and private vehicles through clear, intuitive, and barrier-free connections.

One of the major components of the vision is the transformation of the existing surface parking into a people-oriented public space. By shifting vehicle parking to an underground facility, valuable land on the surface can be utilized for pedestrian plazas, landscaped green spaces, seating areas, commercial developments, and organized transport facilities. This approach places greater emphasis on pedestrians rather than vehicles, creating a more welcoming, accessible, and comfortable environment for all users.

The project also seeks to strengthen urban mobility by introducing a dedicated City Bus Terminal that is fully integrated with the existing railway and metro systems. This integration will make transfers between different modes of transport faster and more convenient while helping to reduce traffic congestion around the station. Dedicated zones for taxis, auto-rickshaws, ride-hailing services, and passenger pick-up and drop-off will further improve traffic management and ensure smoother circulation within the station precinct.

Drawing inspiration from successful international transport hubs such as Kowloon Station, the proposed design envisions a multi-level passenger concourse where transportation facilities, retail spaces, restaurants, waiting areas, and public amenities are integrated into a single interconnected environment. Rather than functioning solely as a transit point, the station precinct is envisioned as an active urban destination that supports commercial activity, enhances passenger comfort, and contributes to the social and economic growth of the city.

Sustainability is another important aspect of the project vision. The proposal encourages greater use of public transportation through efficient multimodal connectivity while incorporating environmentally responsible design strategies. Features such as underground parking, electric vehicle charging stations, energy-efficient lighting, rainwater harvesting, solar energy systems, natural daylighting, and extensive landscaping are included to reduce the environmental impact of the development and create a healthier, more sustainable urban environment.

The project also emphasizes universal accessibility to ensure that the transport hub can be used comfortably by people of all ages and abilities. Barrier-free design elements, including elevators, escalators, ramps, tactile paving, accessible parking spaces, universal toilets, and clear wayfinding systems, are incorporated throughout the development so that every passenger can travel safely, independently, and with ease.

Ultimately, this project aims to establish Rani Kamalapati Railway Station as a benchmark for next-generation multimodal transport hubs in India. By integrating transportation, architecture, public spaces, sustainability, and commercial development into a single cohesive design, the proposal demonstrates how existing railway stations can be transformed into future-ready urban mobility hubs that improve passenger experience, support economic growth, and contribute to the development of smarter and more connected cities.

Scope of the Project

The scope of this project is to develop a conceptual design for transforming the forecourt of Rani Kamalapati Railway Station into an integrated multimodal transport hub. The proposal focuses on improving passenger mobility, strengthening connectivity between different modes of transport, and making better use of the available land, while preserving the existing railway station and metro infrastructure.

The project is limited to the redevelopment of the station forecourt, which is currently used mainly for surface parking. It does not involve changes to the railway station building, platforms, or railway operations. Instead, the proposal concentrates on reorganizing the surrounding transportation facilities to create a more efficient, safe, and convenient environment for passengers transferring between different modes of transport.

One of the major aspects of the project is the planning and design of an underground parking facility for private vehicles and two-wheelers. Relocating parking below ground frees up valuable surface space that can be converted into pedestrian-friendly public areas. The proposal also includes the development of a dedicated City Bus Terminal to improve integration between local bus services, the railway station, and the metro network, making public transport more accessible and efficient.

Another important component is the design of a multi-level passenger concourse inspired by successful international multimodal transport hubs. This concourse is planned to provide seamless connectivity between the railway station, metro station, city buses, underground parking, taxis, auto-rickshaws, and app-based mobility services. To improve the overall passenger experience, the proposal also incorporates facilities such as retail outlets, food courts, waiting lounges, ticketing counters, information centres, public amenities, and landscaped open spaces.

The project further aims to create a safe, inclusive, and universally accessible transportation environment. Barrier-free design features such as elevators, escalators, ramps, tactile paving, accessible parking spaces, and continuous pedestrian pathways are incorporated to ensure that people of all ages and abilities can move comfortably throughout the station precinct.

Sustainability is also an important part of the proposed development. The design integrates environmentally responsible strategies including rainwater harvesting, solar energy utilization, electric vehicle charging stations, natural daylighting, energy-efficient lighting, and landscape development. These measures promote sustainable urban mobility while reducing the environmental impact of the project.

The scope also includes site analysis, assessment of the existing transportation infrastructure, study of international multimodal transport hubs, conceptual planning, master planning, circulation planning, parking design, space programming, and the preparation of architectural drawings, 3D models, and visualizations to effectively communicate the proposed design.

However, the project does not cover modifications to railway tracks, signalling systems, train operations, metro alignment, or the structural reconstruction of the existing railway station and metro station buildings. The proposal is conceptual in nature and is intended to demonstrate how the strategic redevelopment of the station forecourt can improve multimodal connectivity, enhance passenger experience, and transform Rani Kamalapati Railway Station into a future-ready transportation hub while making the best use of the existing infrastructure.

INTERNATIONAL CASE STUDY

Kowloon Station, Hong Kong



Introduction

Kowloon Station, Hong Kong, is internationally recognized as one of the most successful examples of a multimodal transport hub. Developed as part of Hong Kong's integrated railway network, the station demonstrates how transportation infrastructure can be combined with commercial development, public spaces, and urban planning to create a highly efficient and passenger-oriented mobility ecosystem.

Unlike conventional railway stations that primarily function as boarding and alighting points, Kowloon Station has evolved into a mixed-use urban destination where transportation, retail, commercial activities, and public life coexist within a single integrated environment. The station serves as an important interchange between metro services, airport rail, buses, taxis, pedestrians, and private vehicles while maintaining clear passenger circulation and high operational efficiency.

The planning principles adopted at Kowloon Station have become a global benchmark for designing modern multimodal transport hubs, particularly in dense urban environments where land availability is limited and passenger volumes are continuously increasing.

Project Information

Project Name: Kowloon Station

Location: Kowloon, Hong Kong

Transit Operator: MTR Corporation

Type: Multimodal Transport Hub

Integrated Transportation Modes

- Metro Rail (MTR)
- Airport Express
- Urban Railway
- Public Bus Services
- Taxi Services
- Pedestrian Network
- Private Vehicle Access
- Underground Parking

Major Components

- Passenger Concourse
- Commercial Retail
- Restaurants and Food Courts
- Waiting Areas
- Vertical Circulation Cores
- Underground Parking
- Multiple Entry and Exit Points
- Public Open Spaces

Why Kowloon Station was Selected

Kowloon Station was selected as the primary international case study because it successfully demonstrates how multiple transportation systems can function as one integrated mobility hub without compromising passenger comfort or operational efficiency.

The station effectively combines transportation infrastructure with commercial development, public amenities, and pedestrian-oriented urban design. Rather than separating different transport modes, Kowloon Station organizes them through a hierarchy of circulation spaces that enable passengers to move quickly and intuitively between rail services, metro, buses, taxis, and surrounding developments.

Another significant feature of the station is its emphasis on vertical integration. Different transportation modes are distributed across multiple levels and connected through elevators, escalators, and staircases, reducing conflicts between pedestrian and vehicular movement while maximizing land utilization.

The station also demonstrates how commercial spaces can be strategically integrated along passenger circulation routes, creating active public environments that improve passenger experience while generating additional revenue to support station operations.

Planning Principles Observed

The analysis of Kowloon Station identified several planning strategies that contribute to its success as a world-class multimodal transport hub.

- **Seamless Multimodal Integration**

All transportation modes are connected through short, intuitive, and weather-protected transfer routes, significantly reducing passenger transfer time.

- **Multi-Level Planning**

Transportation facilities are organized vertically across different levels, allowing efficient separation of pedestrian movement, vehicle circulation, parking, and public transport operations.

- **Passenger-Oriented Circulation**

Passenger movement follows direct and clearly identifiable routes supported by effective wayfinding systems and strategically positioned vertical circulation cores.

- **Integrated Commercial Development**

Retail outlets, restaurants, and public amenities are positioned along major circulation corridors, transforming waiting and transfer spaces into vibrant urban environments.

- **Efficient Land Utilization**

Underground parking and stacked transportation facilities maximize the use of limited urban land while preserving valuable surface areas for pedestrians and public activities.

- **Universal Accessibility**

The station incorporates elevators, escalators, ramps, tactile paving, and barrier-free circulation, ensuring accessibility for passengers of all age groups and physical abilities.

Relevance to the Proposed Project

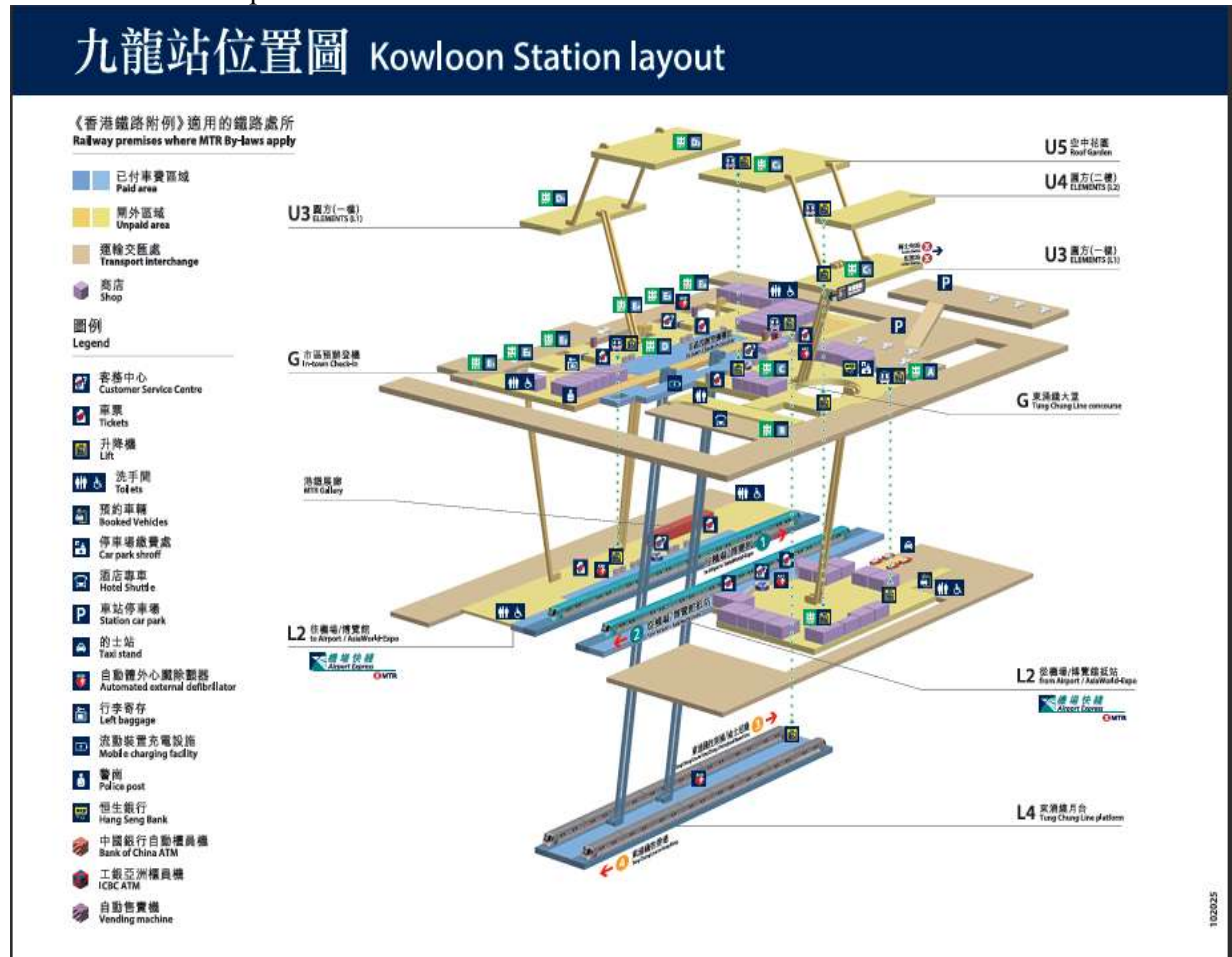
Although Rani Kamalapati Railway Station differs significantly from Kowloon Station in terms of scale, passenger volume, and urban context, several planning principles can be successfully adapted to improve multimodal connectivity.

The proposed redevelopment does not attempt to replicate Kowloon Station. Instead, it selectively adopts design concepts such as integrated passenger circulation, underground

parking, transit-oriented retail, public plazas, organized multimodal transfers, and efficient vertical circulation.

These principles have been carefully modified to suit the physical constraints and transportation requirements of Rani Kamalapati Railway Station while preserving its existing railway and metro infrastructure.

The case study demonstrates that successful multimodal transport hubs are not defined solely by advanced transportation systems but by the quality of passenger movement, integration between different modes of transport, and the creation of active public spaces that enhance the overall travel experience.

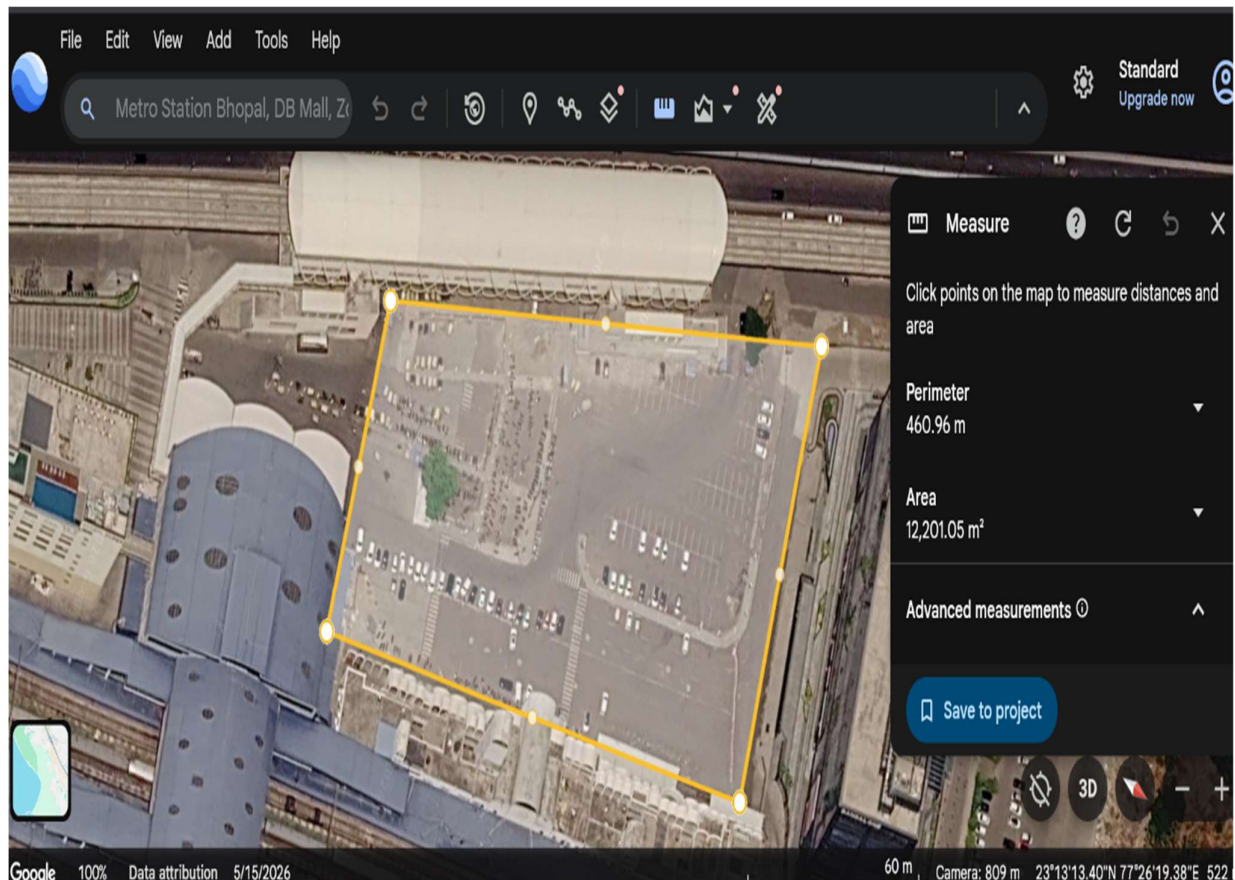


Current Project Progress

The initial planning phase of the project has been successfully completed. This included site analysis, study of the existing transportation infrastructure, review of international case studies, concept development, and preliminary space planning. Based on the finalized concept, the project has now moved into the Building Information Modeling (BIM) stage using Autodesk Revit.

Currently, the development of the underground parking level (Level -1) is in progress. The layout has been planned to accommodate organized parking for private cars and two-wheelers while ensuring smooth internal circulation. It includes dedicated entry and exit ramps, internal roadways, lift and escalator cores, pedestrian access routes, EV charging stations, and utility spaces required for efficient operation.

The underground parking forms the foundation of the proposed multimodal transport hub. By relocating vehicle parking below ground, the existing surface parking area can be transformed into a spacious pedestrian plaza with integrated transport facilities. This approach not only improves land utilization but also supports the project's vision of creating a safe, efficient, and passenger-friendly multimodal transport hub at Rani Kamalapati Railway Station.

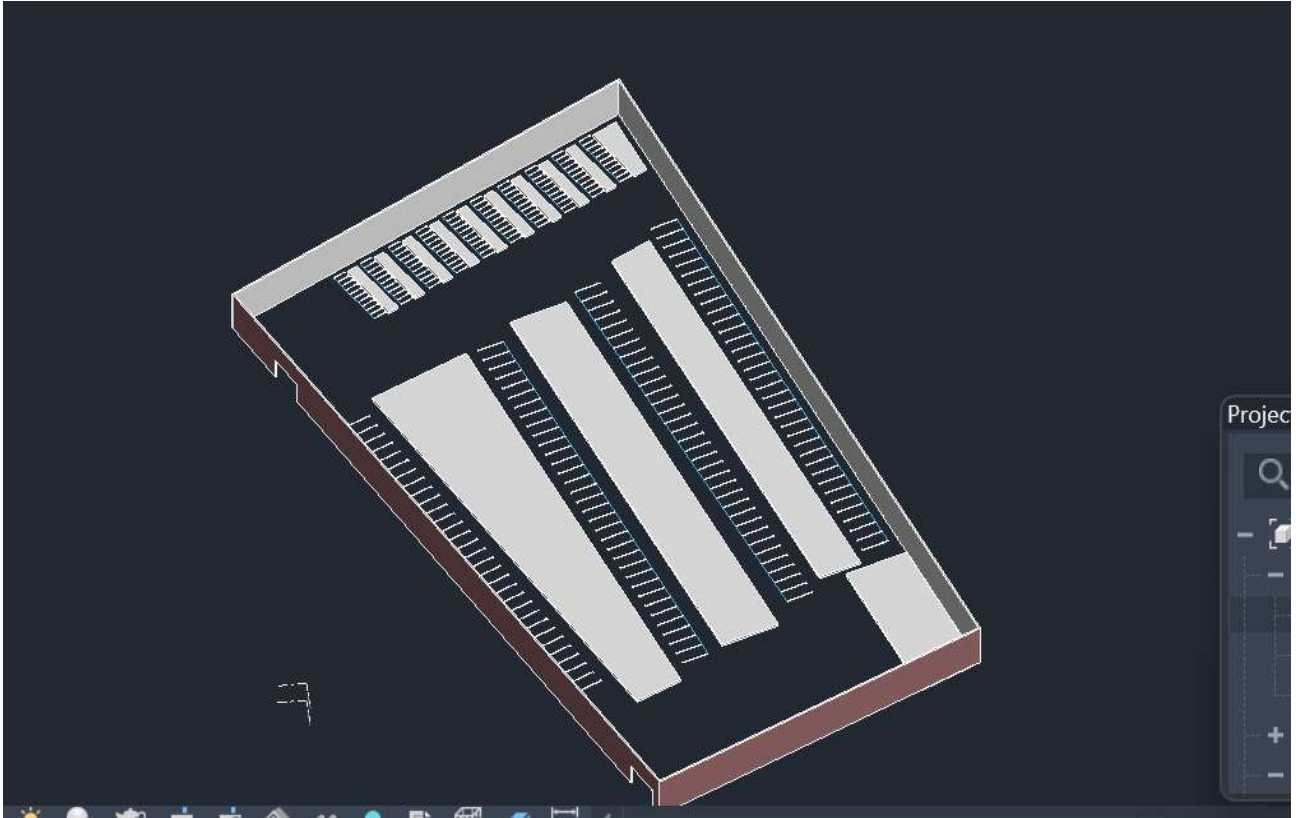


The underground parking level (Level -1) is currently under development. The parking layout has been planned to accommodate organized parking for private cars and two-wheelers, with dedicated circulation aisles, separate entry and exit ramps, pedestrian movement corridors, and service areas. The basement design also incorporates EV charging stations, designated accessible parking spaces, utility rooms, fire safety provisions, and two passenger lifts that provide direct vertical connectivity to the passenger plaza and station concourse.

In parallel, the design team is developing passenger-centric facilities that will enhance the overall user experience. These include dedicated health and first-aid facilities, emergency response spaces, EV charging infrastructure, organized taxi and auto-rickshaw circulation, and universally accessible movement through elevators, ramps, and barrier-free pathways.

The objective is to create a safe, efficient, and sustainable mobility hub that prioritizes passenger comfort while ensuring seamless integration between all modes of transport.

The next phase of the project includes the development of the ground-level passenger plaza, City Bus Terminal, integrated retail concourse, metro connectivity, landscape planning, and three-dimensional visualization of the complete multimodal transport hub.



Sustainable Infrastructure

Sustainability is one of the key principles guiding the proposed multimodal transport hub. The project has been designed to create an efficient, environmentally responsible, and passenger-friendly transportation system by combining renewable energy, sustainable resource management, and modern mobility solutions. Every design decision aims to reduce the environmental footprint while enhancing the overall travel experience.

To meet a significant portion of the station's energy requirements, the proposal includes a **500 kW rooftop solar photovoltaic system** consisting of approximately **900 solar panels**. The electricity generated will be used to power common facilities such as lighting, elevators, escalators, passenger waiting areas, and other public amenities, reducing dependence on conventional energy sources and lowering operational costs.

Efficient water management is another important aspect of the project. A **rainwater harvesting system** with an estimated storage capacity of **500,000 litres** is proposed to collect and store rainwater for landscape irrigation, toilet flushing, and general maintenance.

purposes. This will help conserve potable water and improve the station's overall water efficiency.

The project also supports the growing adoption of electric vehicles by providing **20 EV charging stations for cars** and **40 charging points for two-wheelers** within the underground parking facility. In addition, the integration of the City Bus Terminal, organized taxi and auto-rickshaw bays, and well-planned pedestrian pathways encourages the use of public transport and reduces dependency on private vehicles, contributing to lower traffic congestion and reduced carbon emissions.

To create a healthier and more comfortable environment, the proposal includes approximately **3,000 m² of landscaped open spaces** with more than **120 native trees** and extensive green cover. These landscaped areas improve the visual quality of the station, provide shaded public spaces, reduce the urban heat island effect, and contribute to a better microclimate for passengers.

Energy-efficient LED lighting, daylight-responsive lighting controls, and smart building management systems have been incorporated to optimize energy consumption throughout the facility. The proposal also includes waste segregation points, a **300 KLD Sewage Treatment Plant (STP)** for wastewater recycling, and recycling facilities to promote responsible waste management.

Overall, the proposed sustainable infrastructure is intended to create a future-ready transport hub that not only improves mobility but also promotes environmental conservation, resource efficiency, and long-term operational sustainability. By integrating green technologies with passenger-centric design, the project contributes towards the vision of developing smarter, cleaner, and more sustainable urban transportation systems.